

Software Requirements Specification

for

Chitara The AI Music Generator

Version 1.2

Prepared by Yatichapat Dechaeamsakul

Kasetsart University Computer Engineering Department

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the AI Music Generation Web Application. The purpose of this document is to clearly define the system's behavior, features, and constraints to ensure a shared understanding among all stakeholders, including developers, project managers, testers, and the product owner.

This SRS covers the full scope of the AI music generation platform, including user authentication, AI-based song generation, music playback, library management, sharing, and access control. It serves as a reference throughout the system's design, development, testing, and maintenance phases.

1.2 Document Conventions

This document follows standard Software Requirements Specification (SRS) conventions to ensure clarity and consistency across all requirements for the Chithara AI Music Generator.

The following keywords are used to indicate the level of requirement obligation:

Terms	Description
SHALL	Indicates a mandatory requirement. The system must implement this requirement to be considered compliant with this SRS.
SHOULD	Indicates a recommended requirement that is desirable but not mandatory.
MAY	Indicates an optional requirement that can be implemented at the developer's discretion.

Requirements are categorized as follows:

Requirement Number	Description
US-XX	Use Story identifier, where XX represents a unique number.
UC-XX	Use Case Model, where XX represents a unique number.
FR-XX	Functional Requirement identifier, where XX represents a unique number.

NFR-XX	Non-Functional Requirement identifier, where XX represents a unique number.
BR-XX	Business Role identifier, where XX represents a unique number.

1.3 Intended Audience and Reading Suggestions

This document is intended for the stakeholders listed in Table 2. Each stakeholder group may focus on different sections of this SRS based on their role and interest in the system.

Stakeholder	Role
AI music creator	People who use the app to generate, listen to, organize and share music
Shared link user	People who receive shared music links
Product Owner / Client	Represents business goals and customer needs

Reading Suggestions

Readers are encouraged to begin with Section 1 (Introduction) to gain an overview of the system's purpose and scope.

AI Music Creators may focus on the User Stories, Functional Requirements, and System Features sections to understand how the system supports music creation, playback, and sharing.

Shared Link Users may focus on the Sharing and Access Control and Error Handling requirements to understand how shared content can be accessed.

Product Owner / Client may focus on the Product Scope, System Features, and Non-Functional Requirements sections to ensure alignment with business objectives and constraints.

This document is organized to allow readers to easily locate information relevant to their role while maintaining a complete description of the system.

1.4 Product Scope

Chithara is a web-based AI music generation application that allows users to create personalized music using artificial intelligence. Users can specify song attributes such as title, genre, occasion, mood, and description to generate unique AI-generated songs.

The system provides features for music playback, automatic storage of generated songs, album organization, downloading for offline use, and

controlled sharing through public, invite-only, or private access. Users authenticate via Google accounts to ensure secure access and ownership of generated content.

The primary objective of Chithara is to support creative music generation in an easy-to-use web platform while maintaining secure access control and a smooth user experience.

1.5 References

- IEEE 29148-2018, "Systems and Software Engineering — Life Cycle Processes — Requirements Engineering."
- Google Identity Services Documentation, Google OAuth 2.0: <https://developers.google.com/identity/protocols/oauth2?hl=th>
- Software Engineering Course Project Specification, "Cithai Music Generator" Scenario
- AI Music Generation API Documentation (Suno API).

2. Overall Description

2.1 Product Perspective

Chithara is a web-based application that provides AI-assisted music generation and management capabilities. The system allows users to generate music by supplying structured input parameters, including title, genre, occasion, mood, and description. These inputs are processed by an AI music generation service to produce audio content.

The application includes features for music playback, automatic storage of generated songs, album organization, and controlled sharing. These features exist to support common user workflows such as revisiting previously generated content, organizing multiple songs, and providing access to selected users through shareable links.

User authentication is handled through Google accounts to uniquely identify users and associate generated content with a specific account. Privacy and access control mechanisms are included to restrict or allow access to generated music based on user-defined visibility settings.

From the user's perspective, the system reduces the technical effort required to generate and manage music by providing a single web interface that integrates generation, storage, playback, and sharing functionalities.

2.2 Product Functions

Major functions

- Authenticate users using Google accounts
- Accept structured input for music generation, including title, genre, occasion, mood, and description
- Generate music using an AI music generation service after user confirmation
- Display generation progress and completion status
- Automatically store generated songs in a personal library
- Allow users to play, pause, rewind, fast-forward, and skip music tracks
- Enable users to organize songs into albums

- Allow users to search and filter songs based on metadata
- Allow users to download generated songs
- Support sharing of music via links with configurable access control
- Handle invalid links and system errors gracefully

Out of Scope

- No administrative dashboard or admin management system will be provided.
- No user-to-user direct messaging or social networking features will be included.
- No native mobile application (iOS/Android) will be developed in this version.
- No integration with external music streaming platforms (e.g., Spotify, Apple Music) will be included.

2.3 Stakeholders

Product Owner

- The Product Owner defines the business objectives, system scope, and project constraints. This stakeholder is responsible for approving requirements and ensuring that the system meets functional, budgetary, and timeline expectations. The Product Owner does not directly interact with the system during normal operation.

End Users (AI Music Generator)

- End Users are the primary stakeholders who use the system to generate, manage, and share AI-generated music. They are concerned with usability, music quality, generation speed, reliability, and data privacy. Their feedback directly influences system improvements and feature prioritization.

Shared Link Receivers

- Shared Link Users access music content through shared URLs. They interact only with limited playback functionality. Their primary concerns are ease of access, playback reliability, and clear feedback in case of invalid or restricted links.

2.4 User Classes and Characteristics

The system supports the following user classes:

AI Music Creator

This is the primary user class. AI Music Creators use the system to generate, manage, and share AI-generated music. These users are expected to have basic experience with web applications but do not require technical knowledge of AI or music production systems. They have full access to system features, including music generation, playback, organization, downloading, and sharing.

Shared Link User

Shared Link Users are secondary users who access music content via a shared link provided by an AI Music Creator. Their access is limited strictly to the specific content associated with the shared link.

2.5 Operating Environment

The system operates in a web-based environment with the following characteristics:

Client-Side Environment

- The application is accessible through modern web browsers, including Chrome, Safari, Firefox, and Edge.
- The user interface is designed to support desktop devices running common operating systems.

External Integrations

- User authentication is provided through Google Identity Services (Google OAuth 2.0).
- Music generation functionality integrates with the Suno AI music generation service. The system sends structured song parameters to the Suno API and receives generated audio files in response.

2.6 Design and Implementation Constraints

The development and implementation of the Chithara AI Music Generator are subject to the following constraints:

Technical Constraints

- The system shall be implemented as a web-based application and accessed through standard web browsers.
- User authentication shall be performed using Google OAuth 2.0 through Google Identity Services.
- Music generation shall rely on a third-party AI music generation API, and the system must comply with the API's usage limitations and response formats.
- Generated audio files shall be handled in commonly supported audio formats (e.g., MP3).

Resource Constraints

- The system shall operate within the limitations of free or low-cost tiers of third-party AI music generation services.

Tools and Framework Constraints

- The application shall be developed using standard web development technologies compatible with modern browsers.
- The database system shall support scalable storage of user data and music metadata.

Security and Policy Constraints

- The system shall enforce access control rules based on user-defined privacy settings.
- User data and generated content shall be protected in accordance with standard web security practices.
- Access to private and invite-only music shall be restricted to authorized users only.

2.7 Assumptions and Dependencies

Assumptions

- Users have access to a stable internet connection when using the system.
- Users access the system using a supported modern web browser.
- The database system provides reliable data persistence and integrity.
- The Suno AI music generation service remains operational and accessible.

- The Suno API maintains a stable request and response format during the system's deployment period.
- Generated audio files are compatible with standard web audio playback formats.
- Sufficient server-side storage capacity is available for storing generated audio files.

Dependencies

- The system depends on Google Identity Services (OAuth 2.0) for user authentication and identity verification.
- The system depends on the Suno AI music generation API to generate audio content.
- The system relies on backend hosting infrastructure for processing requests and storing data.
- The system relies on standard web technologies and browser support for user interaction and audio playback.

3. Stakeholders

Stakeholder	Role Descriptions
End User (AI music creator)	End Users are the primary stakeholders who use the system to generate, manage, and share AI-generated music. They are concerned with usability, music quality, generation speed, reliability, and data privacy. Their feedback directly influences system improvements and feature prioritization.
Shared link user	Shared Link Users access music content through shared URLs. They interact only with limited playback functionality. Their primary concerns are ease of access, playback reliability, and clear feedback in case of invalid or restricted links.
Product Owner / Client	The Product Owner defines the business objectives, system scope, and project constraints. This stakeholder is responsible for approving requirements and ensuring that the system meets functional, budgetary, and timeline expectations. The Product Owner does not directly interact with the system during normal operation.

4. User Stories

4.1 AI Music Creator

- **US-01:** As an AI Music Creator, I want to log in using a Google account so that I can securely access my account and system features.
- **US-02:** As an AI Music Creator, I want to enter song details including title, genre, occasion, mood, and description so that I can generate personalized music based on my input.
- **US-03:** As an AI Music Creator, I want to confirm the song details before generation so that I can verify the information.
- **US-04:** As an AI Music Creator, I want to see a visual loading indicator during song generation so that I know the system is processing my request.
- **US-05:** As an AI Music Creator, I want generated songs to be automatically saved in my personal library so that I do not lose my content.
- **US-06:** As an AI Music Creator, I want to create albums and add selected songs to them so that I can organize my music.
- **US-07:** As an AI Music Creator, I want to search and filter songs by genre, creation date, or occasion so that I can quickly find relevant music.
- **US-08:** As an AI Music Creator, I want to play and pause songs using a web-based player so that I can control playback.
- **US-09:** As an AI Music Creator, I want to fast-forward or rewind a song by a fixed duration (e.g., 10–15 seconds) so that I can skip to specific parts.
- **US-10:** As an AI Music Creator, I want to skip to the next or previous song so that I can navigate between tracks.

- **US-11:** As an AI Music Creator, I want to download generated songs so that I can listen to them offline.
- **US-12:** As an AI Music Creator, I want to share music using a link so that others can access it.
- **US-13:** As an AI Music Creator, I want to set my music to public so that anyone with the link can access it.
- **US-14:** As an AI Music Creator, I want to set my music to invite-only so that only authorized email addresses can access it.
- **US-15:** As an AI Music Creator, I want to set my music to private so that only I can access it.
- **US-16:** As an AI Music Creator, I want to see my remaining generation quota so that I can manage my usage effectively.
- **US-17:** As an AI Music Creator, I want the system to handle errors such as invalid or broken links gracefully so that the application does not crash.

4.2 Shared Link User

- **US-18:** As a Shared Link User, I want to access shared music via a link so that I can listen to the content easily.
- **US-19:** As a Shared Link User, I want to see a clear message if access is denied or the link is invalid so that I understand the issue.

5. External Interface Requirements

5.1 User Interfaces

The system provides a web-based graphical user interface (GUI) that allows users to interact with all system functions through a standard web browser.

The user interface includes the following logical components:

Authentication Interface:

- A login interface that allows users to authenticate using their Google account.

Music Generation Interface:

- A form-based interface where users input song details such as title, genre, occasion, mood, and description, followed by a confirmation step before generation.

Library and Album Interface:

- Interfaces that allow users to view their personal music library, create albums, add or remove songs, and view album contents.

Music Playback Interface:

- A web-based music player that provides controls such as play, pause, rewind, fast-forward, and track navigation.

Search and Filter Interface:

- Controls for searching and filtering songs based on metadata such as genre, creation date, or occasion.

Sharing Interface:

- Interfaces that allow users to generate shareable links and configure privacy settings for music content.

Error and Feedback Messages:

- The system displays clear messages for loading states, successful operations, and error conditions such as invalid or expired links.

5.2 Software Interfaces

The system interacts with the following external software components:

Google Identity Services (OAuth 2.0)

- Used for user authentication and identity management.

AI Music Generation Service

- A third-party AI service used to generate music based on user-provided input parameters. The system sends structured song data and receives generated audio files in response. In this project, using Suno AI as the instructor recommendation

Data Storage System

- Used to store user accounts, generated songs, albums, metadata, and access control information.
 - Data exchanged between components includes authentication tokens, song metadata, audio files, and access permissions.
- Communication between software components follows standard API-based interaction models.

6. Functional Requirements

This section describes the major system features provided by Chithara and the associated functional requirements.

6.1 User Authentication

6.1.1 Description

This feature allows users to authenticate using a Google account to access system functionality and associate generated music with a unique user identity.

6.1.2 Functional Requirements

- FR-01: The system shall allow users to log in using a Google account.
- FR-02: The system shall uniquely identify users based on their Google account.
- FR-03: The system shall allow shared link receivers to authenticate using Google accounts when required.

6.2 AI Music Generation

6.2.1 Description

This feature allows users to generate music by providing structured song information and confirming the generation request. The system interacts with an AI music generation service and provides feedback during the generation process.

6.2.2 Functional Requirements

- FR-04: The system shall allow users to input song details before generating.
- FR-05: The system shall allow users to review and edit song details before confirming generation.
- FR-06: The system shall generate a song only after the user confirms the input details.
- FR-07: The system shall display a loading indicator while a song is being generated.
- FR-08: The system shall notify the user when the song generation is completed or failed.
- FR-09: The system shall track the number of songs the user has generated.

6.3 Music Library and Album Management

6.3.1 Description

This feature enables users to store, view, and organize generated songs in a personal library and albums.

6.3.2 Functional Requirements

- FR-10: The system shall automatically store the generated song into the user's personal library immediately after generation.
- FR-11: The system shall allow users to view all their previously generated songs in their personal library.
- FR-12: The system shall allow users to add selected songs into their album.

6.4 Music Playback and Download

6.4.1 Description

This feature allows users to listen to generated music and control playback, as well as download songs for offline use.

6.4.2 Functional Requirements

- FR-13: The system shall allow users to play songs from their library.
- FR-14: The system shall provide playback controls including play and pause.
- FR-15: The system shall allow users to fast forward or rewind the song by a fixed duration.
- FR-16: The system shall allow users to skip to the next or previous song.
- FR-17: The system shall allow users to download generated songs for offline use.

6.5 Search, Sharing, and Access Control

6.5.1 Description

This feature enables users to search for music content and control access through sharing and privacy settings.

6.5.2 Functional Requirements

- FR-18: The system shall allow users to search and filter songs by title, genre, and creation date.
- FR-19: The system shall allow users to set a song or album to private.
- FR-20: The system shall allow users to set a song or album to public via a shareable link.
- FR-21: The system shall allow users to restrict access to invite-only users via email.

7. Other Nonfunctional Requirements

7.1 Performance Requirements

The system shall meet the following performance requirements under normal operating conditions:

- NFR-1: The system shall respond to user interface actions such as button clicks, navigation, and form submissions within 2 seconds under normal network conditions.
- NFR-2: The system shall begin music playback within 3 seconds after a user selects a song, provided the audio file is available and network conditions are stable.

7.2 Safety Requirements

The Chithara AI Music Generator does not control physical hardware or operate in environments where physical harm could occur. However, the system shall implement safeguards to prevent loss of user data and unintended system behavior.

- NFR-3: The system shall provide confirmation or validation before performing actions that may result in data loss, such as deleting songs or removing content from albums.
- NFR-4: The system shall display clear error messages when operations fail, so that users are aware of issues and can take appropriate action.
- NFR-5: The system shall ensure that invalid or broken shared links do not expose unintended content or cause system instability.

7.3 Security Requirements

The system shall ensure the protection of user accounts, generated music, and shared content through appropriate authentication, authorization, and secure communication mechanisms.

- NFR-11: The system shall authenticate users using Google OAuth to verify user identity before allowing access to protected features.
- NFR-12: The system shall protect authentication tokens and user session information from unauthorized exposure or misuse.
- NFR-13: The system shall restrict access to invite-only shared content to users whose email addresses have been authorized.
- NFR-14: The system shall prevent unauthorized modification or deletion of songs, albums, and user data.

7.4 Software Quality Attributes

The system shall satisfy the following software quality attributes:

- Usability: The system shall provide a clear and consistent user interface so that users can generate and manage music without requiring prior technical knowledge.
- Reliability: The system shall handle errors gracefully and display appropriate error messages instead of crashing.
- Maintainability: The system shall be designed with modular components to allow future improvements or feature additions.

7.5 Business Rules

The following business rules apply to the operation of the Chithara AI Music Generator:

- BR-01: Only authenticated users may generate songs and manage personal libraries.
- BR-02: Generated songs are associated with the user who created them and can only be modified or deleted by that user.
- BR-03: Access to shared songs or albums shall follow the selected privacy settings (public, invite-only, or private).
- BR-04: Shared link users may only view or play content that has been explicitly shared with them.

8. Other Requirements

- Generated songs shall be stored and delivered in a standard audio format supported by modern web browsers (e.g., MP3 or WAV).

9. Use Case Model

UC-1: Log in with Google

Primary Actor: AI Music Creator

Preconditions: The user has a valid Google account.

Postconditions: The user is authenticated and the user session is created.

Main Success Scenario:

1. The user selects "Log in with Google."
2. The system redirects the user to Google authentication.
3. The user enters Google credentials.
4. Google verifies the user identity.
5. The system receives the authentication token.
6. The system creates or retrieves the user account.
7. The user is redirected to the personal dashboard or music personal library.

UC-2: Generate Song

Primary Actor: AI Music Creator

Preconditions: The user is registered into the system.

Postconditions: A generated song is stored in the user's personal library and the user is notified of a successful generation.

Main Success Scenario:

1. The user enters song details (title, genre, mood, occasion, description).

2. The system displays a review page.
3. The user confirms the song details.
4. The system sends the request to the AI music generation service.
5. The system displays a loading indicator.
6. The AI service returns a generated audio file.
7. The system stores the song in the user's personal library.
8. The system notifies the user that generation is successful.

UC-3: Manage Library

Primary Actor: AI Music Creator

Preconditions: The user is logged into the system and has at least one generated song in the user's account.

Postconditions: The user's library or album structure is updated. Any changes made (e.g., album creation, song deletion, privacy update) are saved in the system.

Main Success Scenario:

1. The user navigates to the Personal Library page.
2. The system displays all previously generated songs.
3. The user selects a management action (e.g., create album, edit album, delete song, set privacy level, share album).
4. The system displays the appropriate management interface.
5. The user performs the selected action.
6. The system validates the action.
7. The system updates the library data.
8. The system confirms the successful update to the user.

UC-4: Share Music

Primary Actor: AI Music Creator

Preconditions: The user is logged into the system and has at least one song exists in the user's library.

Postconditions: A shareable link is generated and the selected privacy setting is applied.

Main Success Scenario:

1. The user selects a song or album.
2. The user chooses "Share."

3. The user selects privacy level (public, invite-only, or private).
4. If invite-only, the user enters authorized email addresses.
5. The system generates a shareable link.
6. The system stores access control settings.
7. The user copies and distributes the link.

UC-5: Access Shared Music

Primary Actor: Shared Link User

Preconditions: A valid shareable link exists. The user has a valid Google account and the system is accessible via a web browser.

Postconditions: The user is authenticated. The shared music content is displayed and the user is able to play the music if authorized.

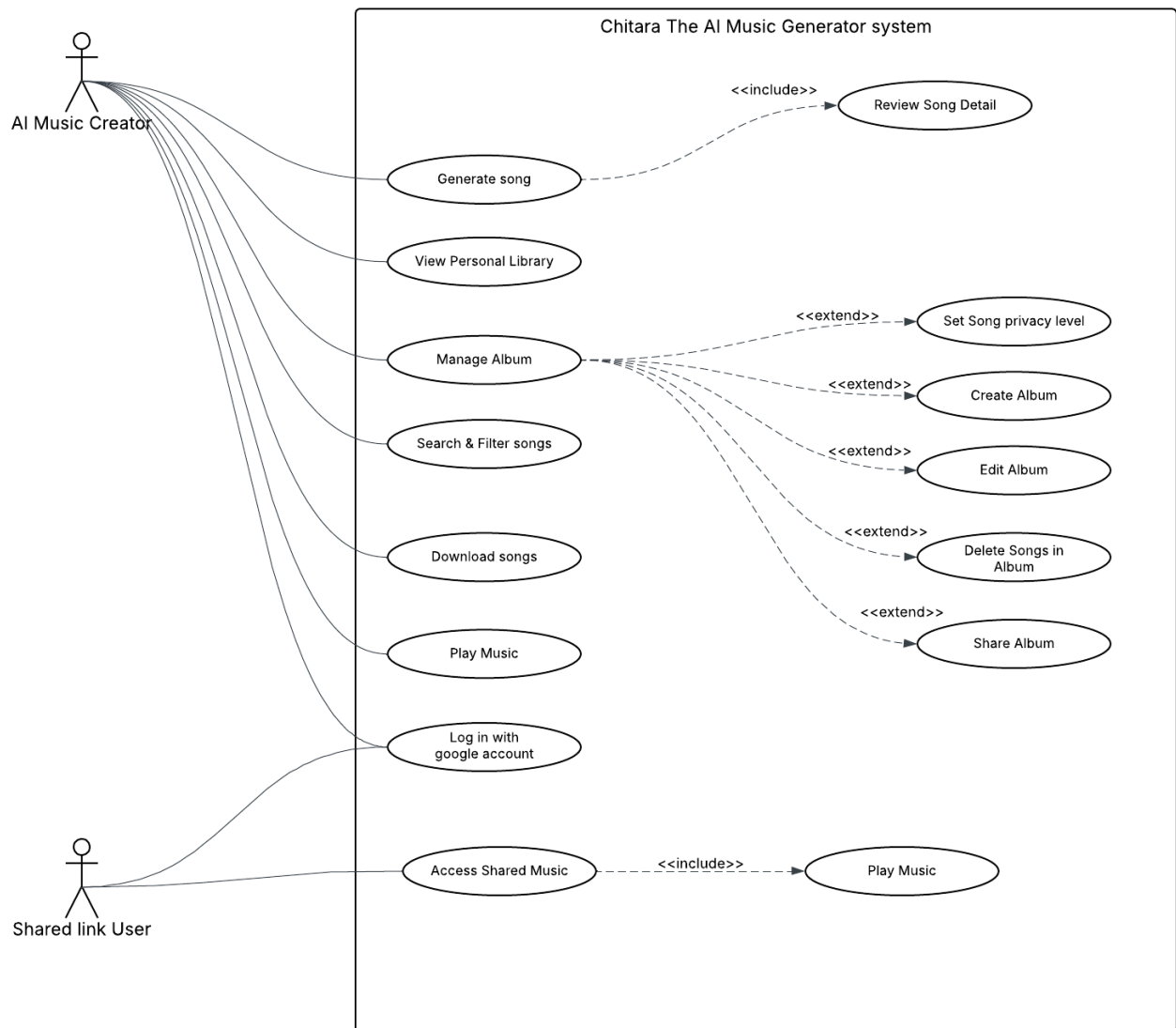
Main Success Scenario:

1. The user opens a shared link.
2. The system prompts the user to log in with a Google account.
3. The user authenticates via Google.
4. The system verifies the authentication token.
5. The system validates the user's access permission for the shared content.
6. The system displays the shared music page.
7. The user plays the music.

Appendix A: Glossary

Term	Definition
AI	Artificial Intelligence, used in the system to generate music based on user input.
Authentication	The process of verifying a user's identity before granting access.
Album	A collection of songs organized by a user.
Library	The personal collection of songs generated by a user.
Shareable Link	A URL that allows authorized users to access shared music content.
Privacy Settings	Access control levels such as Public, Invite-Only, or Private that determine who can access music.
Google OAuth 2.0	An authentication method provided by Google to verify user identity securely.

Appendix B: Analysis Models



Use Case Model Summary

The Chithara AI Music Generator system involves two primary actors: AI Music Creator and Shared Link User.

AI Music Creator is the primary actor who interacts with the system to generate and manage AI-generated music. They can:

- Log in with Google account to securely access system features.
- Generate Song, which includes reviewing song details before submission.
- View Personal Library to see previously generated songs.
- Manage Album, which includes:
 - Setting song privacy level
 - Creating an album
 - Editing an album

- Deleting songs from an album
- Sharing an album
- Search & Filter Songs based on metadata.
- Download Songs for offline use.
- Play Music using the built-in web player.
- The Generate Song use case includes reviewing song details before final confirmation.
- The Manage Album use case includes several sub-use cases related to album organization and privacy control.

Shared Link User is a secondary actor who accesses content shared by the AI Music Creator. They can:

- Access Shared Music through a shareable link.
- Play the shared content (included in the access process).